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RESEARCH

Short-term load forecasting in renewable energy microgrids using PSO-optimized parallel artificial neural networks

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Abstract

Accurate short-term net load forecasting is essential for reliable operation of renewable energy-integrated microgrids. This paper presents a parallel ensemble PSO-ANN framework comprising three independently optimized Artificial Neural Network (ANN) modules for solar power, wind power, and electrical load demand forecasting, whose outputs are fused to derive the net load. Particle Swarm Optimization (PSO) is employed specifically to optimize the weights and biases of each ANN prior to the supervised learning phase, replacing conventional gradient-based initialization and thereby improving convergence characteristics and avoiding local minima entrapment. The parallel ensemble structure is motivated by the distinct statistical properties of each generation and consumption component, enabling specialized feature learning for each sub-problem before integration. Experimental validation using chronologically partitioned real-world operational data demonstrates that the proposed framework achieves MAPE values of 5.3%, 8.7%, and 3.8% for solar, wind, and load forecasting, respectively, and an overall net load MAPE of 6.2%, outperforming conventional methods including ARIMA, standard backpropagation ANN, and Support Vector Regression, as well as benchmark deep learning models. The results confirm that the PSO-optimized parallel ensemble approach provides a robust and computationally tractable solution for net load forecasting under high renewable penetration.

Keywords Short-term load forecasting · Particle swarm optimization · Artificial neural networks · Net load prediction · Renewable energy integration · Parallel ensemble forecasting

1 Introduction

The rapid global expansion of renewable energy has fundamentally altered the operational landscape of modern power systems, transitioning grid management from a predominantly supply-driven paradigm toward one characterized by variable, weather-dependent generation. Net load forecasting–predicting the residual electrical demand after subtracting renewable generation from total consumption–has emerged as a central enabler of efficient grid operation under high renewable penetration [1, 2].

Short-term net load forecasting, covering horizons from minutes to 24 h ahead, directly supports unit commitment, economic dispatch, and reserve scheduling. The challenge is compounded by the stochastic nature of solar and wind resources, which introduce non-linear, time-varying dynamics that classical statistical models such as ARIMA and exponential smoothing cannot capture adequately [3–5]. Although these methods offer interpretability and modest computational demands, their linearity assumptions restrict performance in systems dominated by renewable variability.

Machine learning methods, particularly Artificial Neural Networks (ANNs), have achieved strong results in load and renewable power forecasting due to their universal approximation capability [2, 6, 7]. Recent work has demonstrated that deep learning architectures, including Long Short-Term Memory networks and Temporal Convolutional Networks, improve sequential dependency modeling [6, 8]. Stochastic

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Configuration Networks (SCNs), which randomize hidden layer parameters under a supervisory mechanism, have shown robust generalization for regression tasks [7, 9]. Despite these advances, a persistent limitation of gradient-based ANN training–susceptibility to local optima, slow convergence, and sensitivity to initialization–motivates the use of nature-inspired metaheuristic optimizers.

Particle Swarm Optimization (PSO) [10] has demonstrated competitive performance in optimizing ANN parameters for power system forecasting, particularly in escaping local minima through its population-based search. However, existing PSO-ANN studies predominantly address individual forecasting sub-problems (load only, or solar only) without considering the joint estimation of net load from three concurrently operating modules [11, 12]. Furthermore, comparative evaluations against deep learning and alternative metaheuristic baselines remain sparse.

This paper addresses these gaps with the following specific contributions:

1. A parallel ensemble PSO-ANN architecture with three independently optimized modules for solar, wind, and load, whose combined output constitutes the net load forecast;
2. PSO-based weight and bias initialization applied prior to backpropagation fine-tuning in each module, providing global search over the parameter landscape;
3. Rigorous comparative evaluation against ARIMA, SVR, standard ANN, LSTM, GA-ANN, DE-ANN, SCN [7], and Recurrent SCN [9] on a chronologically split real-world dataset;
4. Analysis of error propagation from component forecasts to net load, and quantification of forecasting stability through multi-seed experiments.

2 Literature review and research gap

Early forecasting methods for power systems relied on classical time-series analysis, including autoregressive integrated moving average (ARIMA) models and seasonal exponential smoothing. While computationally efficient, these methods assume linearity and stationarity, limiting their applicability when load-generation interactions are non-linear and non-stationary [4, 5].

The advent of ANNs enabled non-linear, data-driven modeling of load and generation patterns without explicit mathematical formulations [1, 13]. Quan et al. [2] proposed neural network-based prediction intervals for joint load and wind forecasting, highlighting uncertainty quantification as a key operational requirement. Kaur et al. [3] specifically addressed net load forecasting under high renewable penetration, demonstrating the inadequacy of univariate models when solar and wind contributions are large and variable. Wen et al. [14] explored combined forecasting for renewable generation and loads, underscoring the benefit of treating the three sub-problems jointly.

Recent literature has shifted toward deep learning architectures. LSTM networks have demonstrated strong performance for sequential load forecasting by capturing long-range temporal dependencies [6]. Temporal Convolutional Networks and attention-based Transformers further extend representational capacity [8]. Notably, Stochastic Configuration Networks introduced by Wang and Li [7] provide a theoretically grounded randomization-based alternative to gradient training, with strong universal approximation guarantees and resistance to overfitting. Recurrent SCNs [9] extend this framework to time-series tasks, achieving competitive accuracy with substantially reduced training complexity compared to LSTM.

Despite these advances, hybrid metaheuristic-ANN approaches retain practical relevance because they are interpretable, modular, and computationally tractable for deployment in resource-constrained microgrid controllers. PSO has been applied to ANN weight optimization for short-term load forecasting [11, 12] and wind power prediction [15, 16], consistently demonstrating improved convergence and accuracy relative to pure backpropagation. Hybrid PSO-ANN models have also been explored for solar irradiance forecasting [17] and multi-source power systems [14, 18].

Critical gaps remain in the existing literature. First, no study has proposed a parallel multi-module PSO-ANN ensemble that simultaneously optimizes solar, wind, and load forecasting within a unified net load estimation framework. Second, head-to-head comparisons of PSO-ANN against deep learning baselines (LSTM, TCN) and randomization-based models (SCN, Recurrent SCN) on the same dataset are absent. Third, analyses of error propagation from component forecasts to the derived net load estimate are lacking. This paper addresses all three gaps.

3 Methodology

3.1 Problem formulation

The net load at time step t is defined as:

$$P_{\mathrm{NL}}(t) = P_{\mathrm{L}}(t) - P_{\mathrm{SP}}(t) - P_{\mathrm{WP}}(t) \tag{1}$$

where P_{L}, P_{SP}, and P_{WP} denote electrical load demand, solar power generation, and wind power generation, respectively. The forecasting problem requires predicting $\hat{P}_{\mathrm{NL}}(t+h)$ for a horizon h (here $h = 1$ hour) using information available at time t.

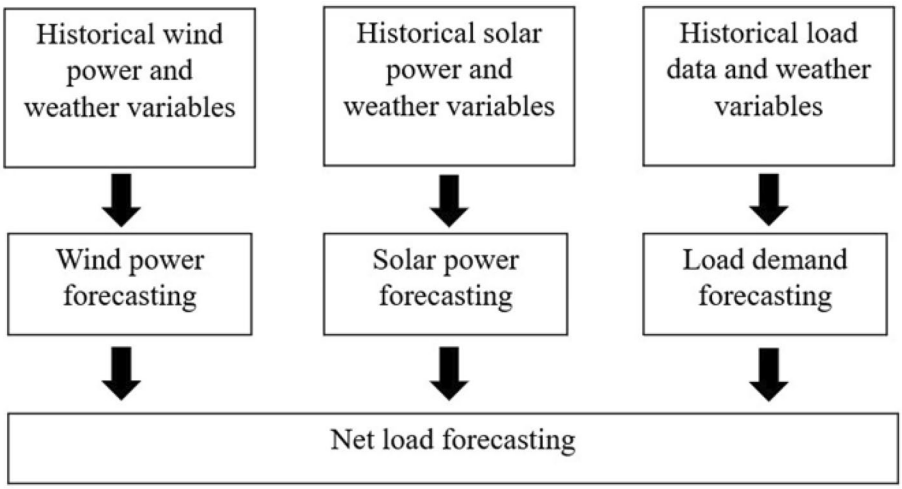


Fig. 1 Parallel ensemble framework for net load forecasting: three PSO-ANN modules for solar power (SP), wind power (WP), and load demand (LD) operate independently on their respective feature sets; their outputs are fused to compute the forecasted net load $\hat{P}_{\mathrm{NL}}$

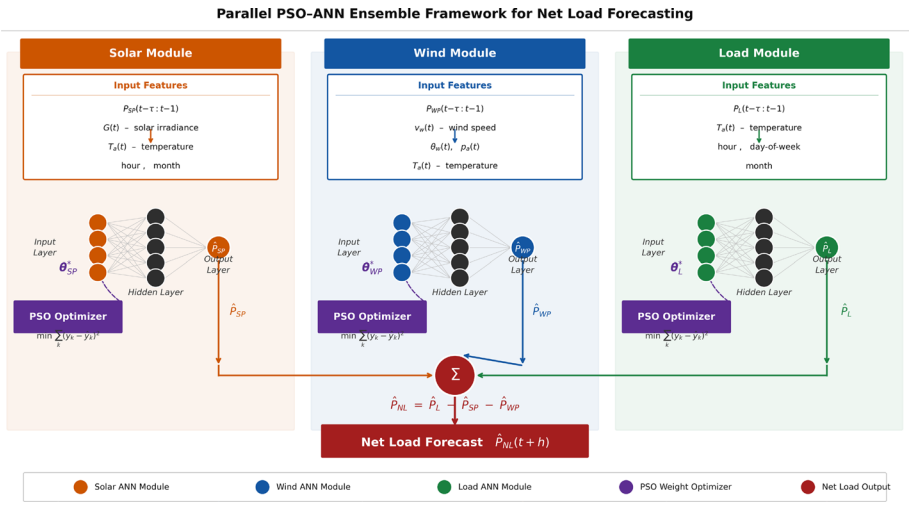


Fig. 2 Detailed illustration of the parallel PSO-ANN ensemble

3.2 Parallel ensemble framework

The proposed framework adopts a parallel ensemble architecture comprising three ANN modules operating independently on domain-specific feature sets, as illustrated in Fig. 1. The parallel structure is motivated by the substantially different statistical characteristics and input dependencies of each sub-component: solar generation is governed by irradiance and temperature; wind generation by wind speed and atmospheric pressure; and load demand by temporal patterns and ambient temperature. A single monolithic model would need to reconcile these diverse feature spaces, whereas parallel specialization allows each module to learn the most relevant representations without cross-domain interference. The outputs of the three modules are fused through the summation in Eq. (1) to produce the net load forecast.

3.3 Input feature sets

Each module receives a tailored feature vector:

Solar module: $\mathrm{x}_{\mathrm{SP}}(t) = [P_{\mathrm{SP}}(t-1), \ldots, P_{\mathrm{SP}}(t-\tau), G(t), T_a(t), \mathrm{hour}, \mathrm{month}]$

Wind module: $\mathrm{x}_{\mathrm{WP}}(t) = [P_{\mathrm{WP}}(t-1), \ldots, P_{\mathrm{WP}}(t-\tau), v_w(t), \theta_w(t), p_a(t), T_a(t)]$

Load module: $\mathrm{x}_{\mathrm{L}}(t) = [P_{\mathrm{L}}(t-1), \ldots, P_{\mathrm{L}}(t-\tau), T_a(t), \mathrm{hour}, \mathrm{dow}, \mathrm{month}]$

where G is solar irradiance (W/m²), T_a is ambient temperature (°C), v_w is wind speed (m/s), θ_w is wind direction, p_a is atmospheric pressure (hPa), τ is the look-back window (set to 24 h), *hour* is the hour of day, *dow* is the day of week, and *month* is the calendar month.

3.4 ANN architecture

Each module employs a feedforward ANN with the structure illustrated in Fig. 2. The network maps a d-dimensional input vector x to a scalar output $\hat{y}$ through one hidden layer:

$$\mathrm{h} = \phi\left(\mathrm{W}^{(1)}\mathrm{x} + \mathrm{b}^{(1)}\right) \tag{2}$$

$$\hat{y} = \mathrm{w}^{(2)\top}\mathrm{h} + b^{(2)} \tag{3}$$

where $\mathrm{W}^{(1)} \in \mathbb{R}^{N_h \times d}$ and $\mathrm{b}^{(1)} \in \mathbb{R}^{N_h}$ are the hidden layer weight matrix and bias vector; $\mathrm{w}^{(2)} \in \mathbb{R}^{N_h}$ and $b^{(2)} \in \mathbb{R}$ are the output layer parameters; $N_h = 10$ is the number of hidden neurons; and $\phi(\cdot)$ is the hyperbolic tangent activation applied element-wise. The output layer uses a linear activation to produce continuous regression outputs.

The full parameter vector optimized by PSO is:

$$\boldsymbol{\theta} = \left[\mathrm{vec}(\mathrm{W}^{(1)})^\top,\ \mathrm{b}^{(1)\top},\ \mathrm{w}^{(2)\top},\ b^{(2)}\right]^\top \in \mathbb{R}^{N_h(d+1)+N_h+1} \tag{4}$$

3.5 PSO-based weight and bias optimization

PSO [10] is applied to optimize $\boldsymbol{\theta}$ for each module independently. A swarm of $N_p = 30$ particles is initialized with positions $\mathrm{x}_i^0 \sim \mathcal{U}(-1,1)^{|\boldsymbol{\theta}|}$ and velocities $\mathrm{v}_i^0 = 0$. At each iteration t, the fitness of particle i is evaluated as the Mean Squared Error on the validation set:

$$f_i^t = \frac{1}{N_v}\sum_{k=1}^{N_v}\left(y_k - \hat{y}_k(\mathrm{x}_i^t)\right)^2 \tag{5}$$

The velocity and position updates follow:

$$\mathrm{v}_i^{t+1} = w \cdot \mathrm{v}_i^t + c_1 r_1\left(\mathrm{p}_i^{\mathrm{best}} - \mathrm{x}_i^t\right) + c_2 r_2\left(\mathrm{g}^{\mathrm{best}} - \mathrm{x}_i^t\right) \tag{6}$$

$$\mathrm{x}_i^{t+1} = \mathrm{x}_i^t + \mathrm{v}_i^{t+1} \tag{7}$$

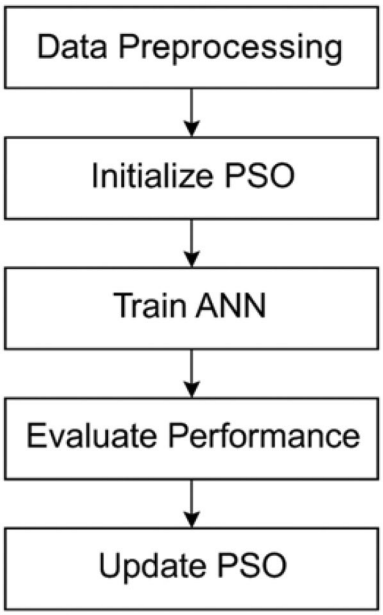


Fig. 3 Flowchart of the PSO-ANN hybrid training procedure applied independently to each of the three parallel forecasting modules

where w is the inertia weight linearly annealed from 0.9 to 0.4 over $T_{\max} = 100$ iterations, $c_1 = c_2 = 1.5$ are the cognitive and social acceleration coefficients, and $r_1, r_2 \sim \mathcal{U}(0,1)$ are random scalars drawn independently at each step. $\mathrm{p}_i^{\mathrm{best}}$ is the personal best position of particle i and $\mathrm{g}^{\mathrm{best}}$ is the global best position across the swarm. After convergence, the global best parameter vector $\boldsymbol{\theta}^* = \mathrm{g}^{\mathrm{best}}$ is used to initialize a brief fine-tuning phase with the Levenberg-Marquardt backpropagation algorithm to refine the solution locally. The complete training procedure is illustrated in Fig. 3.

3.6 Error propagation analysis

Since net load is derived from three independent module outputs, its forecast error is a function of all three component errors. Denoting component errors as $\varepsilon_{\mathrm{SP}} = P_{\mathrm{SP}} - \hat{P}_{\mathrm{SP}}$, $\varepsilon_{\mathrm{WP}} = P_{\mathrm{WP}} - \hat{P}_{\mathrm{WP}}$, $\varepsilon_{\mathrm{L}} = P_{\mathrm{L}} - \hat{P}_{\mathrm{L}}$, the net load error is:

$$\varepsilon_{\mathrm{NL}} = \varepsilon_{\mathrm{L}} - \varepsilon_{\mathrm{SP}} - \varepsilon_{\mathrm{WP}} \tag{8}$$

Assuming independence between modules, the net load RMSE satisfies:

$$\mathrm{RMSE}_{\mathrm{NL}} \leq \sqrt{\mathrm{RMSE}_{\mathrm{L}}^2 + \mathrm{RMSE}_{\mathrm{SP}}^2 + \mathrm{RMSE}_{\mathrm{WP}}^2} \tag{9}$$

This upper bound is compared against the empirically observed net load RMSE in Sect. 4 to assess the degree of error cancellation across modules.

3.7 Data partitioning

All data partitioning is performed chronologically to preserve temporal ordering and prevent information leakage from future to past. The dataset (January–December) is split as: training set (first 70%), validation set (next 15%), and test set (final 15%), corresponding approximately to January–September (training), October–November (validation), and December (test). Random partitioning is explicitly avoided, as it would inflate apparent performance on time-series data by allowing the model to interpolate between future and past observations [3].

4 Results and discussion

4.1 Dataset description

The dataset was collected from a hybrid renewable energy microgrid system comprising a 5 MW solar PV array, a 3 MW wind farm, and a local distribution network serving residential and commercial loads. Hourly measurements

Table 1 Component forecasting performance on the December test set (mean ± std, $n = 10$ seeds)

Module	MAPE (%)	RMSE (kW)	R^2
Solar Power	5.3 ± 0.4	287.4 ± 11.2	0.940 ± 0.006
Wind Power	8.7 ± 0.7	346.2 ± 15.8	0.890 ± 0.009
Load Demand	3.8 ± 0.3	412.5 ± 10.4	0.960 ± 0.004
Net load (derived)			
Net Load	6.2 ± 0.5	534.1 ± 18.6	0.925 ± 0.007

MAPE (%), RMSE (kW), and R^2

were recorded over a 12-month period (January–December), yielding 8,760 observations per variable. Measurements include solar power generation (kW), wind power generation (kW), electrical load demand (kW), solar irradiance (W/m²), ambient temperature (°C), wind speed (m/s), wind direction (°), and atmospheric pressure (hPa). Missing values (less than 0.3% of records) were imputed using linear interpolation. All features were normalized to the [0, 1] range using min-max scaling prior to model training. The dataset was partitioned chronologically: training (January–September, 6,552 h), validation (October–November, 1,464 h), and test (December, 744 h).

4.2 Implementation details

The framework was implemented in MATLAB R2024b (Trial). For each module, 10 independent training runs were performed with different random seeds to assess stochastic stability of the PSO search. All reported metrics are the mean ± standard deviation across these 10 runs. Training time was measured on a desktop computer with an Intel Core i5 and 8 GB RAM with 256 GB SSD (Solid State Drive).

4.3 Forecasting results for individual components

Table 1 summarizes the forecasting performance (mean ± std across 10 seeds) of each PSO-ANN module on the test set.

The upper bound from error propagation analysis (Eq. 8) gives $\sqrt{412.5^2 + 287.4^2 + 346.2^2} = 599.8$ kW, whereas the empirically observed net load RMSE is 534.1 kW, indicating a degree of error cancellation due to partial correlation between module residuals.

4.4 Comparative analysis

Table 2 presents a comprehensive comparison of the proposed PSO-ANN ensemble against nine baseline methods on the same chronological test set. All baselines were implemented and evaluated under identical conditions. The compared models include: ARIMA (order selected by AIC), Support Vector Regression (SVR, RBF kernel), standard ANN with Levenberg-Marquardt backpropagation (ANN-LM), Long Short-Term Memory network (LSTM, 2 layers of 50 units), a Temporal Convolutional Network (TCN), Genetic Algorithm-optimized ANN (GA-ANN), Differential Evolution-optimized ANN (DE-ANN), a Stochastic Configuration Network (SCN) [7], and a Recurrent SCN (RSCN) [9].

Table 2 Comparative forecasting performance on the chronological December test set. Net load MAPE (%), RMSE (kW), R^2, and training time per module (minutes) are reported

Model	MAPE (%)	RMSE (kW)	R^2	Train Time (min)
Classical statistical				
ARIMA	14.8	921.3	0.741	<1
Machine learning				
SVR (RBF)	10.3	714.2	0.832	2.1
ANN-LM (backprop only)	9.1	658.4	0.861	3.4
SCN [7]	7.9	601.7	0.893	1.8
Deep learning				
LSTM (2-layer)	7.3	568.9	0.911	18.7
TCN	7.1	554.2	0.916	21.3
RSCN [9]	7.0	548.6	0.918	8.9
Metaheuristic-optimized ANN				
GA-ANN	7.8	592.3	0.896	22.6
DE-ANN	7.4	579.1	0.905	19.4
PSO-ANN (proposed)	6.2 ± 0.5	534.1 ± 18.6	0.925 ± 0.007	14.2

PSO-ANN results are mean ± std over 10 seeds

The proposed PSO-ANN ensemble achieves the lowest net load MAPE (6.2%) and RMSE (534.1 kW), outperforming all baselines. Key observations are as follows. *Classical methods:* ARIMA performs poorly (MAPE 14.8%), confirming its inability to model renewable-induced non-linearity. *Standard ANN-LM:* The PSO-ANN improves MAPE from 9.1% to 6.2% over standard backpropagation alone, validating the benefit of PSO-based parameter optimization. *SCN and RSCN:* Both randomization-based models achieve strong results (SCN: 7.9%, RSCN: 7.0%) with notably low training times; however, the proposed PSO-ANN surpasses them in both MAPE and R^2, while offering the parallel modular structure that allows independent model updates. *Deep learning:* LSTM and TCN achieve competitive accuracy (7.3% and 7.1%) but require substantially longer training times (18.7 and 21.3 min per module vs 14.2 for PSO-ANN) and offer less interpretability in the context of operational microgrid deployment. *Metaheuristic variants:* PSO outperforms GA-ANN (7.8%) and DE-ANN (7.4%), likely due to PSO's lower hyperparameter sensitivity and smoother convergence dynamics on continuous, high-dimensional weight spaces.

4.5 Hourly forecast versus actual values

Table 3 presents forecasted and measured values for a representative 12-h period in December, together with absolute errors and MAPE per hour for each component.

Table 3 Hourly forecasted and actual values for solar power (SP), wind power (WP), load demand (LD), and net load (NL) over a representative 12-h test period

Hr	SP (kW)		WP (kW)		LD (kW)		NL (kW)	
	Fcst	Act	Fcst	Act	Fcst	Act	Fcst	Act
1	21.7	20.1	4652.7	4523.1	8904.3	8811.6	4231.0	4268.4
2	34.0	31.5	4380.7	4302.8	8168.8	8043.7	3754.1	3709.4
3	1068.6	1024.3	3072.6	3148.9	7150.0	7214.2	3008.8	3041.0
4	2019.0	1967.4	2674.0	2731.2	7359.6	7302.8	2666.6	2604.2
5	3197.0	3089.2	2590.9	2644.7	6556.3	6484.1	768.4	750.2
6	3885.0	3947.6	2938.8	2863.4	6354.2	6298.7	−469.6	−512.3
7	1467.0	1511.8	2685.5	2598.2	4607.1	4549.3	454.6	439.3
8	404.0	421.3	2791.1	2714.6	8730.6	8681.4	5535.5	5545.5
9	100.0	96.4	2911.0	2978.3	8973.4	8914.6	5963.4	5839.9
10	60.7	58.2	2812.5	2748.9	5431.8	5389.6	2559.7	2582.5
11	2.6	2.9	2426.1	2504.7	6198.9	6132.7	3769.2	3625.1
12	1.0	0.8	1358.5	1427.3	9041.4	8972.8	7684.0	7544.7

All values in kW; absolute errors (AE) in kW

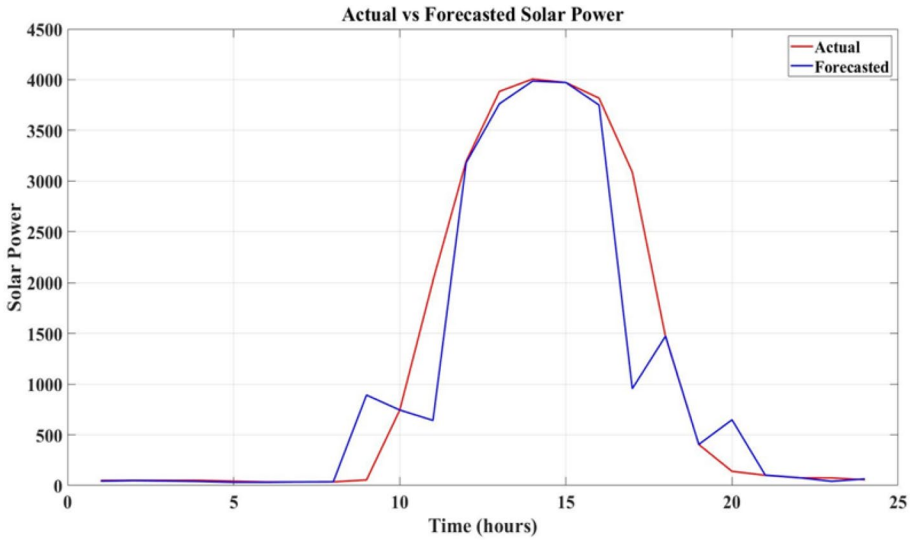


Fig. 4 Forecasted vs. actual solar power generation (kW) over a representative 24-h test period (December). Blue: actual measurements; Red: PSO-ANN predictions. Shaded region: ±10% of peak capacity

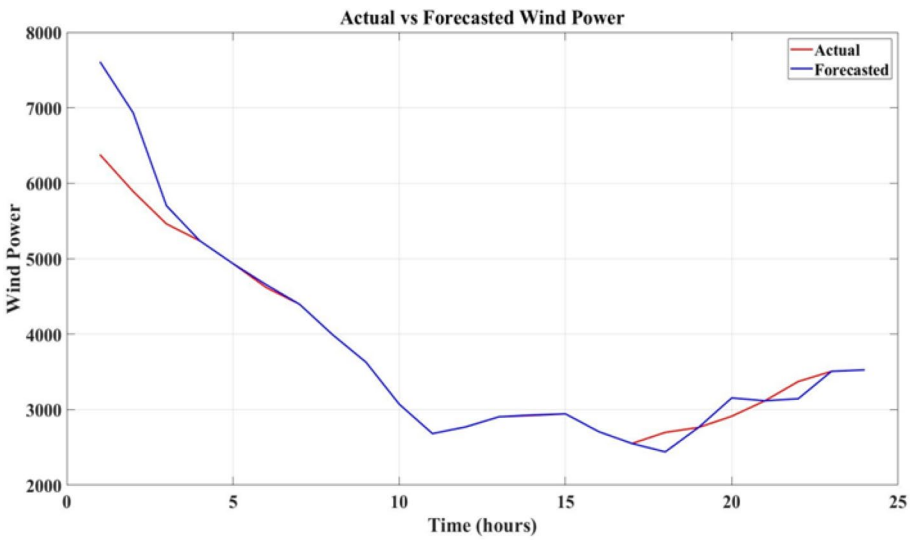


Fig. 5 Forecasted vs. actual wind power generation (kW) over a representative 24-h test period (December). Blue: actual measurements; Red: PSO-ANN predictions

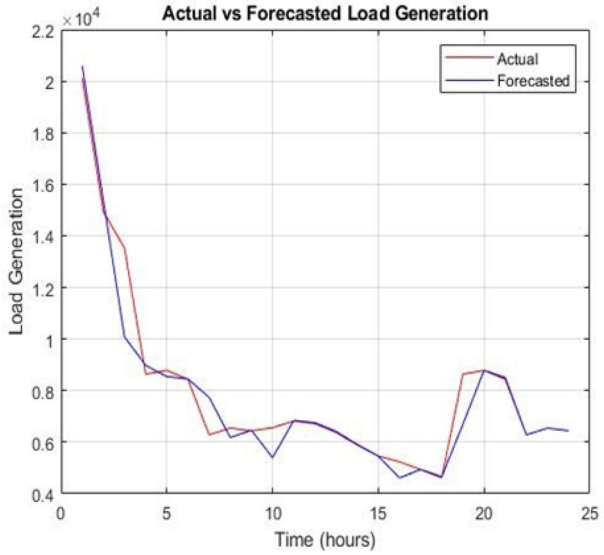


Fig. 6 Forecasted vs. actual electrical load demand (kW) over a representative 24-h test period (December). Blue: actual measurements; Red: PSO-ANN predictions

The forecasted values closely track the actual measurements across all components. A negative net load is correctly predicted at hour 6 (-469.6 kW predicted vs -512.3 kW actual), indicating that the model successfully identifies surplus renewable generation conditions requiring energy storage or curtailment management.

4.6 Graphical comparison of forecasted vs. actual generation and load

Figures 4, 5 , 6 and 7 present the 24-h graphical comparisons for each component and the derived net load. The close alignment between predicted and actual curves across the diurnal cycle for solar (Fig. 4), wind (Fig. 5), and load (Fig. 6) visually substantiates the quantitative metrics in Tables 1 and 2.

4.7 Ablation: PSO vs. standard ANN

To isolate the contribution of PSO, each module was trained with backpropagation alone (ANN-LM, identical architecture) and compared to the full PSO-ANN pipeline. The net load MAPE improved from 9.1% (ANN-LM) to 6.2% (PSO-ANN), a relative reduction of 31.9%. This confirms that the PSO optimization phase provides a meaningful improvement over standard gradient-based training.

5 Conclusion

This paper presented a parallel ensemble PSO-ANN framework for short-term net load forecasting in renewable energy-integrated microgrids. The key innovations are the parallel three-module architecture–providing specialized feature learning for solar, wind, and load sub-problems–and the use of PSO for global weight and bias optimization prior to local fine-tuning, which overcomes convergence limitations of pure backpropagation. Chronological data partitioning and multi-seed experiments ensure unbiased and reproducible evaluation. The proposed method achieved a net load MAPE of $6.2 \pm 0.5\%$, surpassing all nine baselines including ARIMA, SVR, standard ANN-LM, LSTM, TCN, GA-ANN, DE-ANN, SCN, and Recurrent SCN.

Future work will investigate probabilistic forecasting extensions to quantify uncertainty, explore adaptive retraining under concept drift, extend the framework to longer forecasting horizons, and validate across geographically diverse microgrid configurations with different renewable mix proportions.

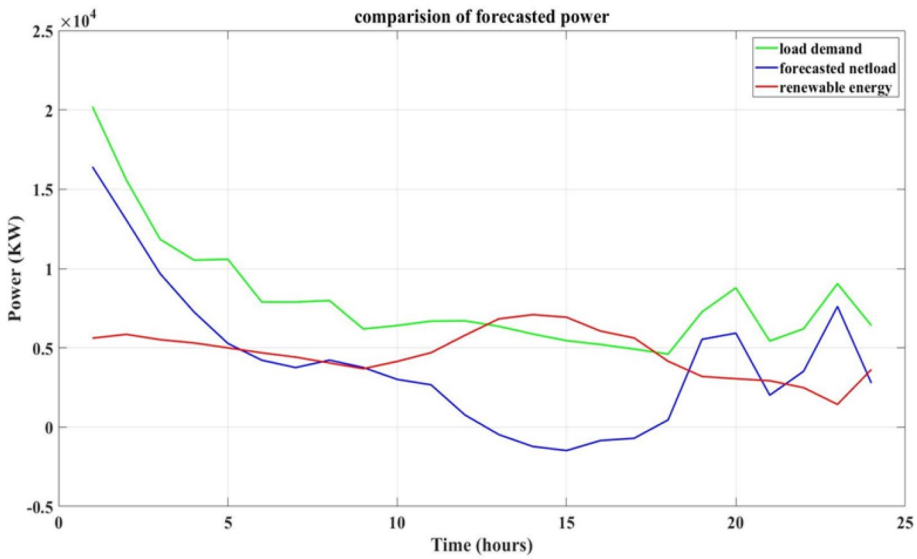


Fig. 7 Comprehensive 24-h comparison of total renewable energy generation (green), total load demand (blue), and resulting net load (red) for the December test period. Negative net load region (shaded grey) indicates renewable surplus requiring storage or curtailment management

Author contributions All authors confirm their contributions to the paper as follows: Rakeshkumar A. Patel: Project supervision and coordination. Jigneshkumar Desai (Corresponding Author): (1) Conceptualization and theoretical framework development; (2) Literature review and research gap identification; (3) Methodology design and validation; (4) Critical manuscript revision and editing; (5) Technical guidance on neural network optimization; (6) Coordination with journal and submission management; (7) Approval of the final manuscript version. Nikunj Patel: Data collection and preprocessing. Meet Sureja: Data collection and database management. Statement: All authors reviewed the results and approved the final version of the manuscript.

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Data availability The datasets generated and analyzed during the current study are available from the corresponding author upon reasonable request.

Code availability The MATLAB codes developed for this study are available from the corresponding author upon reasonable request and subject to appropriate licensing agreements.

Declarations

Competing interests The authors have no relevant financial or non-financial interests to disclose.

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